Home > Regression Analysis > 50 Startups Feature Progression

50 Startups Multiple Linear Regression Explorer

CRISP-DM

Style Ref: StatsUnlock

Feature Selection & Parsimony Analysi

Compare 5 Machine Learning Feature Selection Algorithms **ElasticNet**, **Exhaustive Search**) against Baseline OLS. Explc across feature counts $k=1 \dots 5$, pinpointing the **2-featu 5-feature overfitting cliff**.

50 Startups Dataset | 5 Selection Methods | Global Sv

Overfitting Warning: k=5 ($9,137.99 RMSE)

1. SELECT FEATURE SELECTION ALGORITHM

- Lasso (L1 Ranking) — Matches Benchmark
- Forward SFS — Greedy
- RFE — Backward
- ElasticNet — L1 + L2
- Exhaustive Search — Best Subset
- Compare All 5 Methods

2. NUMBER OF FEATURES (K = 1 TO 5)

- k = 1 (R&D Spend)
- k = 2 ★ Optimal Sweet Spot (Min RMSE)
- k = 3 (+ State / Admin)
- k = 4 (4 Predictors)
- k = 5 ⚠ Full Model (Overfitting Cliff)

TEST RMSE — OPTIMAL SWEET SPOT

$8,198.80

Lower is better (Best across methods)

R² SCORE — 94.7% FIT

0.9474

Explains 94.74% of profit variance

ADJUSTED R²

0.9324

Penalizes redundant parameters

MEAN ABSOLUTE ERROR

$6,886.59

F-STATISTIC — P < 0.0001

349.0

Average prediction error ($)

Model is highly statistically significant

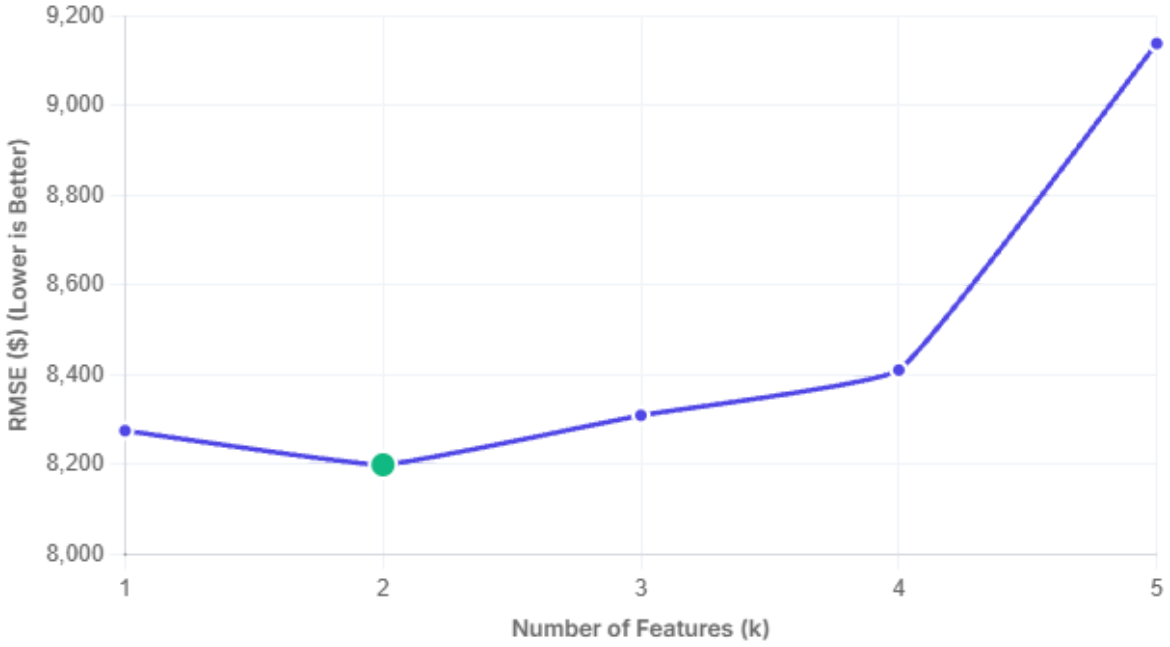
Active Predictors (2): R&D Spend Marketing Spend

Occam's Razor: 2 features yield the minimum test RMSE ($8,198.80) and peak R^2 (0.9474).

RMSE by Number of Features

Lasso

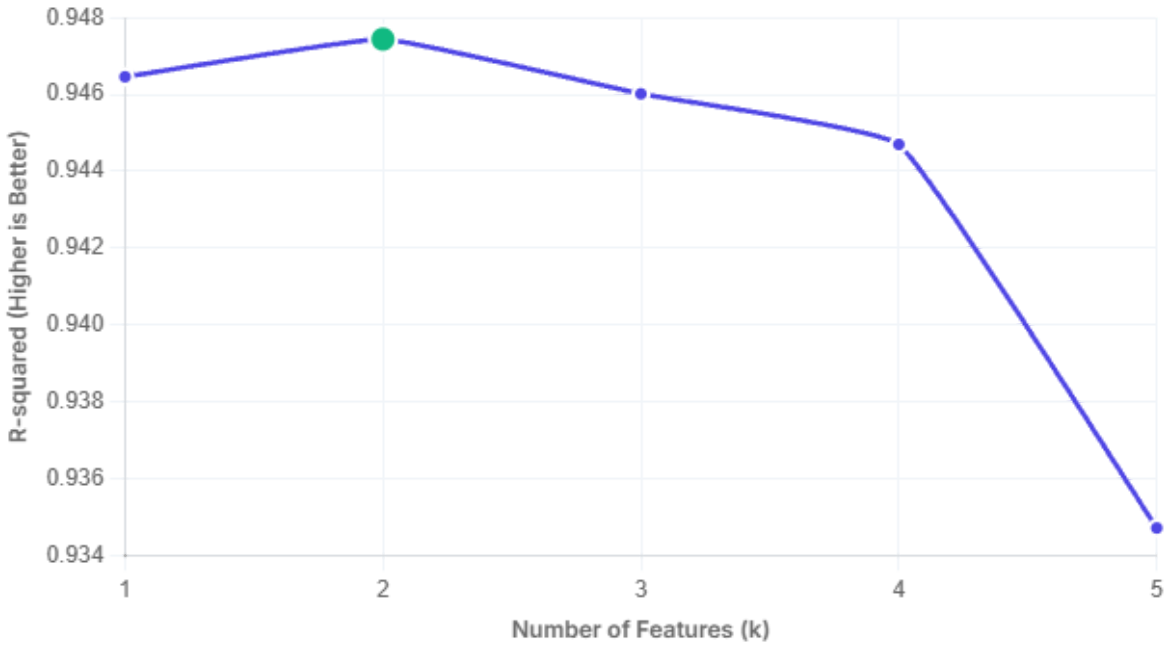
Lower is better. Sharp valley at k=2, jump at k=5.



R-squared by Number of Features

Peak at k=2

Higher is better. Peaks at k=2 (0.9474), drops at k=5.



√× FITTED OLS REGRESSION EQUATION

Profit = 46,346.34 + 0.7886 × (R&D Spend) + 0.0326 × (Marketing Spend)

Model Coefficients & Statistical Tests

Detailed parameter estimates, standard errors, t-statistics, and VIF multicollinearity.

PREDICTOR	ESTIMATE (B)	STD. ERROR	T-STAT	P-VALUE	95% CONF. INT	VIF
Intercept (β₀)	46,346.3388	2971.2365	15.598	< 0.0001 ✓	[40326.04, 52366.64]	-
R&D Spend	0.7886	0.0468	16.846	< 0.0001 ✓	[0.69, 0.88]	2.10 (Safe)
Marketing Spend	0.0326	0.0175	1.860	0.0708 (n.s.)	[-0.00, 0.07]	2.10 (Safe)

Live Profit Predictor

Real-time

Test predictions in real-time with the active model.

R&D Spend ($) $100,000

100000

Marketing Spend ($) $150,000

150000

Administration ($) $120,000

120000

State Location

California (Reference)

PREDICTED STARTUP PROFIT

$130,094.66

Active predictors: R&D Spend, Marketing Spend

(R&D: +$78862 | Marketing: +$4886)

Test Set Residuals & Outliers Diagnostic

Sorted by |Residual|

Top absolute residuals on the out-of-sample test set (10 startups).

RANK	SAMPLE ID	ACTUAL PROFIT	PREDICTED PROFIT	RESIDUAL ($)	ABS ERROR	% ERR
#1	Startup #38	$81,229.06	$68,335.08	+$12,893.98	$12,893.98	15.8
#2	Startup #10	$146,121.95	$134,182.15	+$11,939.80	$11,939.80	8.17
#3	Startup #2	$191,050.39	$180,642.25	+$10,408.14	$10,408.14	5.45
#4	Startup #11	$144,259.40	$133,873.92	+$10,385.48	$10,385.48	7.20

RANK	SAMPLE ID	ACTUAL PROFIT	PREDICTED PROFIT	RESIDUAL ($)	ABS ERROR	% ERR
#5	Startup #27	$105,008.31	$114,717.25	$-9,708.94	$9,708.94	9.25
#6	Startup #22	$110,352.25	$114,580.92	$-4,228.67	$4,228.67	3.83
#7	Startup #4	$166,187.94	$170,343.32	$-4,155.38	$4,155.38	2.50
#8	Startup #41	$77,798.83	$73,701.11	+$4,097.72	$4,097.72	5.27
#9	Startup #28	$103,282.38	$102,284.65	+$997.73	$997.73	0.97
#10	Startup #31	$97,483.56	$97,433.46	+$50.10	$50.10	0.05

🏛 50 Startups CRISP-DM Machine Learning Project

Developed for multiple linear regression model selection, parsimony validation, and statistical inference.
Built with pure Vanilla HTML, CSS, JavaScript, and Chart.js.

REFERENCES & ACKNOWLEDGMENTS

Design & Layout Reference:
Statistical calculator style inspired by StatsUnlock Multiple Linear Regression Calculator (featuring ANOVA, coefficients, VIF, and APA reporting aesthetics).
Dataset: 50 Startups Dataset (Avik Jain 100-Days-Of-ML).
Methodology: CRISP-DM (Cross-Industry Standard Process for Data Mining).